# 6 Supplementary Material

## 6.1 Detailed Metrics Description

Some performance metrics are derived by integrating information from multiple modalities.

*Metric 1: Critical Event Reaction Time.* Critical Event Reaction Time is defined as the duration between the onset of a critical event and the time at which the team begins physically interacting with the corresponding equipment or object to resolve the event. Thus, Reaction Time is measured as the interval between the event onset time and the start of the physical interaction. We use video-derived logs to identify the start time of each critical event (onset time). For the two events analyzed in this paper, event onset is identified based on the corresponding ProPaq and IV equipment logs.

*Metric 2: Critical Event Resolution Time.* Critical Event Resolution Time is measured as the interval between the event onset and event offset. In other words, it represents the duration from the beginning of the critical event until the event is resolved. The corresponding timestamps are derived from equipment logs.

*Metric 3: Task Role Switching Rate.* Task role switching represents team members supporting the team and completing required tasks through role switching and assisting as needed. A team member may perform tasks outside their primary role in response to operational demands; for example, an MD may perform tasks associated with an RN or RT, or vice versa. These behaviors are identified using video-based human–object interaction recognition and trainee role classification. While the RT primarily works with the mechanical ventilator (MV), the RT can also interact with ProPaq to gather information. The nurse primarily works with IV equipment and ProPaq. The MD's primary role is to gather patient status and guide the team. Role-switching rate = number of role-switching episodes / elapsed time from simulation start to Event 2 resolution.

## 6.2 Multimodal Pipeline for Predicting Observable Trainee Actions

*6.2.1 Video-Based Human–Object Interaction Detection.* The video-based human–object interaction (HOI) recognition pipeline builds upon HALDA [1, 6] to identify trainee–equipment interactions. The model produces frame-level interaction confidence scores. Temporal post-processing is then applied to convert these frame-level predictions into temporally localized interaction intervals with defined onset and offset times. The confidence threshold and Gaussian smoothing parameter $\sigma$ were selected through grid search using the validation data, with macro-F1 used to evaluate candidate configurations.

Brief gaps may occur between consecutive frame-level predictions during a continuous physical interaction due to natural pauses or temporary lack of observable interaction. Following the expert annotation schema, gaps of 4.0 seconds or less were accepted as part of the same interaction (natural pause) and merged into a single interaction interval, whereas gaps greater than 4.0 seconds were treated as separate interaction intervals. Predictions from the two wide-angle camera views were then combined to capture all trainee–equipment interactions across the simulation environment.

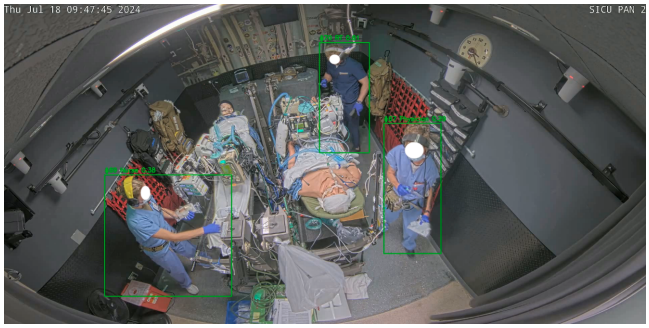



**Figure 3: YOLO-based Person Role Classification (Source: AFRL & University of Cincinnati, Cincinnati, Ohio, USA)**

The video-based HOI detection model achieved a macro-F1 of 83.00% and a weighted F1 of 96.07% on the test data. The model achieved a temporal overlap ratio [7] of 70.00%, with an average of 3.1 false interaction segments [7] per simulation and a false interaction duration [7] of 1.05%.

*6.2.2 YOLO-Based Trainee Role Classification.* To support role-specific trainee-equipment (human-object) interaction analysis, we fine-tuned a YOLO model for trainee detection and role classification using an active-learning-based iterative training procedure. The model detects trainees and classifies their CCATT roles based on color-coded headsets: yellow for nurse, gray for respiratory therapist, and black for physician. During iterative training, model predictions were manually reviewed and corrected, and the verified annotations were incorporated into subsequent training iterations. We applied basic data augmentation techniques, including face blurring, person masking, image blurring, and cropping, to improve robustness to occlusion, blur, and viewpoint variability. Because the headset occupies a small fraction of the trainee bounding box, we also incorporated a contrastive loss to encourage role discrimination and reduce role confusion. The final model achieved an mAP@0.5 of 0.928 and a class-level F1 score of 0.90.

## 6.3 Preliminary evaluation across two critical events and four CCATT teams.

Figure 4 illustrates the temporal alignment of trainee–equipment interactions with the two critical events. The visualization shows interactions across the three trainee roles together with event onset and resolution timings, illustrating how the temporally aligned observations support the derivation of the Reaction Time, Resolution Time, and Task Role Switching Rate metrics.

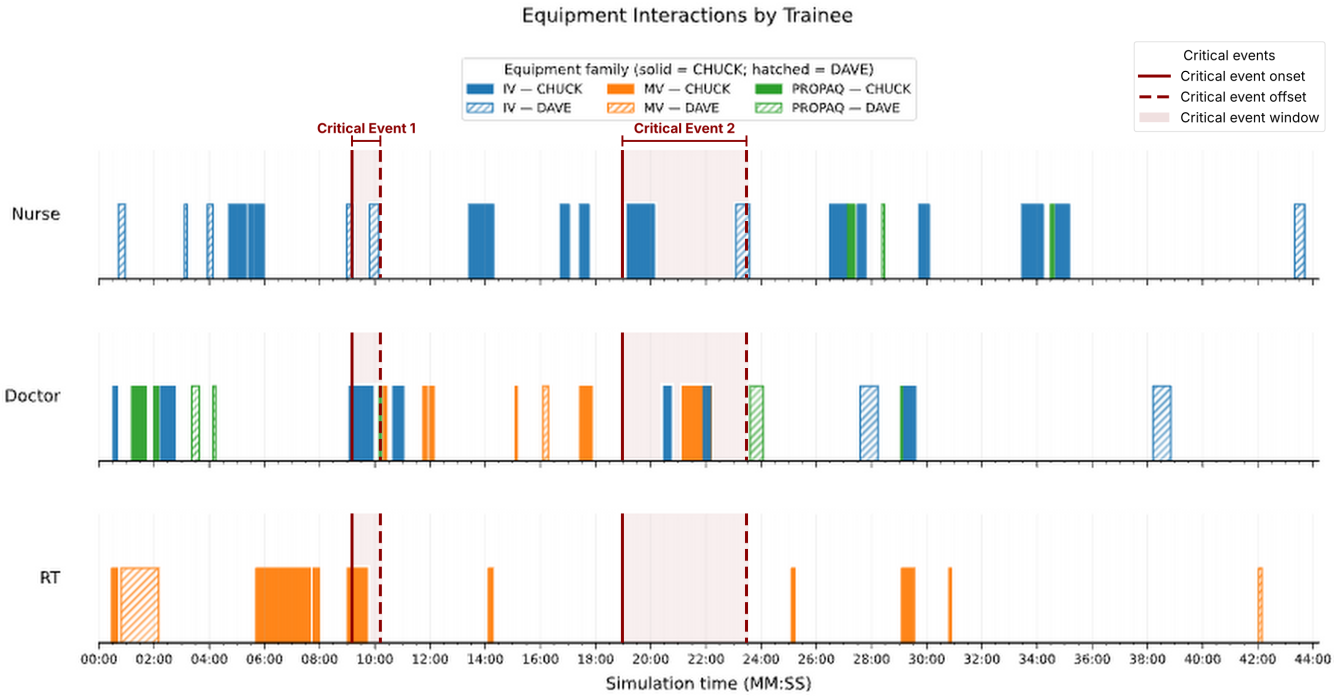



**Figure 4: Multimodal Observable Actions Visualization**

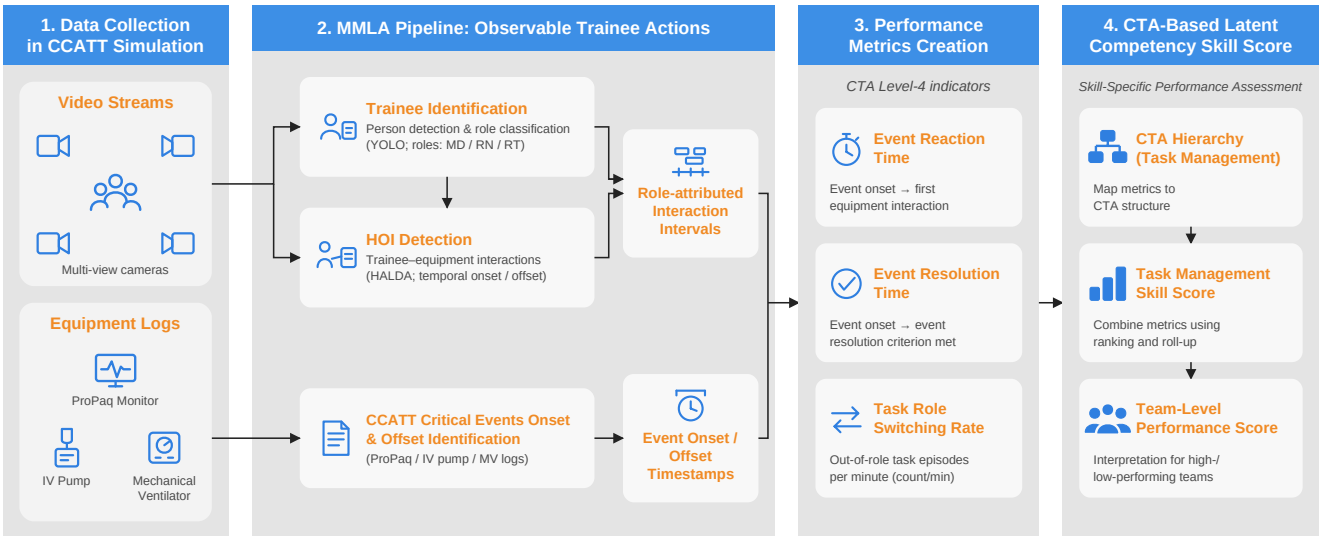



**Figure 5: MMLA–CTA Framework for Automated Trainee Evaluation**